

Binance Automated Trading System (Telegram Integration)

A complete Python-based automated trading system that listens to Telegram signal channels (using `Telethon`), parses signals via Regex, calculates dynamic position sizing for risk management, and executes trades with Bracket (OCO) orders on the Binance API.

Architecture

- `telegram_listener.py` : Background daemon that connects to Telegram using your API Hash, tracking specific channel IDs.
- `signal_parser.py` : Smart NLP/Regex parser to map Forex signals (XAUUSD) to Crypto (BTCUSDT) and extract entry, TP, and SL.
- `risk_manager.py` : Calculates lots sizing dynamically to enforce 1% risk per trade and maximum daily loss guardrails.
- `execution_engine.py` : Implements the trading logic, placing Market Entries followed by OCO Exits (Take Profit + Stop Loss pairs).
- `app.py` : Streamlit graphical dashboard to monitor current equity, open trades, and win-rates.

Installation & Setup

1. Clone & Install Dependencies

```
pip install -r requirements.txt
```

2. Configure Environment Rename `.env.example` to `.env`. Add your Binance API Keys (ensure Testnet is enabled for safety) and Telegram API credentials (from `my.telegram.org`).

3. Running Locally (Without Docker) Open two terminals: *Terminal 1 (Start the Bot)*:

```
python telegram_listener.py
```

Terminal 2 (Start Dashboard):

```
streamlit run app.py
```

4. Running with Docker Compose (Production)

```
docker-compose up -d --build
```

Access the dashboard at `http://localhost:8501`.

Important Security Warning

This bot supports executing live market orders. Keep `.env` out of version control. The bot maps Forex symbols (like XAUUSD) to BTC by default to allow testing Binance's Crypto pairs with Forex-heavy signal groups.